

Proposed Title: Numerical Analysis of Orbital Chaos: Simulating the Restricted Three-Body Problem

Participants:

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Category: 1-Undergraduate (UG) **Subtheme:** Orbital dynamics(planets/satellites)

Abstract:

The classical two-body problem yields elegant, predictable analytical solutions, serving as the historical foundation of celestial mechanics. However, introducing a third body fundamentally alters the system, leading to complex, non-linear dynamics highly sensitive to initial conditions. This project explores the Circular Restricted Three-Body Problem (CR3BP), a theoretical model where a massless third body (such as a satellite or asteroid) moves under the gravitational influence of two massive primary bodies (such as a star and a planet) orbiting their common barycenter.

The primary objective is to computationally model and analyze the chaotic trajectories of the third body to understand regions of orbital stability and instability. Because the CR3BP lacks a general analytical solution, the study will rely heavily on numerical methods. The system's coupled ordinary differential equations (ODEs) will be solved using Python. Moving beyond basic numerical integration methods, a 4th-order Runge-Kutta (RK4) algorithm will be implemented to ensure the necessary precision over extended time steps and limit cumulative computational errors.

The expected outcome includes developing a robust computational simulation capable of visualizing the third body's trajectory in a rotating reference frame. The project will generate phase space plots and potentially Poincare sections to map the boundaries between stable orbits and chaotic behavior. By visualizing these complex orbital dynamics, the project will demonstrate the power of modern computational tools in unlocking the unpredictable nature of multi-body gravitational systems, aligning with the competition's goal of fostering simulation-based analysis in physics.